

A configurable generative transcriptomics framework identifies tissue-dependent synthetic utility in mouse spaceflight RNA-seq

Benchmarking conditional WGAN and diffusion models across NASA OSDR mouse transcriptomes

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Research manuscript draft for author review. Author list, acknowledgments, repository release URL, and archival DOI require final review before submission.

Abstract

Background: Mouse spaceflight studies provide access to tissues that cannot be sampled extensively from astronauts, but individual experiments are small and differ in design. We developed a configurable generative transcriptomics framework and asked whether synthetic expression could improve organ-specific flight analysis without treating generated profiles as new animals.

Methods: We assembled 1,610 mouse flight and ground-control bulk RNA-seq profiles through the NASA Open Science Data Repository API and audited all 997,515 profiles in ARCHS4 mouse. The framework varied expression transformations, feature spaces, harmonization, study scope, tissue structure, conditioning, and training regime. Paper-based WGAN-GP and diffusion implementations were compared using fidelity, real-versus-synthetic separability, distributional-distance, and FLT/GC-effect metrics; GeneJEPA was evaluated as a representation model because it has no expression decoder. The diffusion model used downstream was pretrained on 17,244 tissue-diverse ARCHS4 profiles. Downstream analysis compared pooled and tissue-specific uses of generated expression. Flight associations were tested with real profiles using accession-level random-effects models and Benjamini-Hochberg false-discovery control.

Results: None of nine matched liver harmonization arms provided adequate fidelity and conditional-effect recovery together. A calibrated study-conditioned WGAN-GP achieved correlation 0.976, precision 0.976, recall 0.994, and F1 0.985 on validation, but its adversarial accuracy was 0.636 and accession-aware muscle-effect recovery was 0/6. Across four diffusion seeds, correlation was 0.974, precision 0.997, recall 0.996, F1 0.997, adversarial accuracy was 0.475, the Frechet-distance ratio was 0.074, and muscle-effect recovery was 4/4. Diffusion was therefore used downstream. Pooled augmentation reduced balanced accuracy from 0.754 with real-only training to 0.737 with real-plus-generated training. Tissue-specific use was more informative: 49 real-data BH-FDR associations also entered stable synthetic-informed selection, including a flight-lower mitotic program in thymus and a soleus program with lower *Bdh1*, *Ech1*, *Bnip3*, and *Decr1* and higher *Tpm1*.

Conclusions: Model choice depended on joint fidelity and biological-effect recovery rather than one favorable metric. Diffusion provided the strongest distributionally validated generator, but its downstream value depended on tissue and mode of use. Synthetic expression worked better as a feature prior or regularizer than as additional biological sample size. The thymus, soleus, kidney, spleen, skin, and pooled skeletal-muscle results define testable hypotheses that require independent biological replication.

Keywords: spaceflight; bulk RNA-seq; synthetic data; diffusion model; skeletal muscle; thymus; NASA OSDR; ARCHS4

Introduction

Spaceflight affects immune, musculoskeletal, metabolic, and barrier tissues through a combination of microgravity, radiation, confinement, altered nutrition, stress, and mission-specific procedures. Mouse flight experiments provide tissue access that is unavailable in astronauts, but their transcriptomic interpretation is difficult. Individual studies are small, missions differ in strain and duration, and condition labels can be entangled with study, material, genotype, or collection protocol. Pooling samples without preserving these design variables can convert study effects into apparent flight biology.

The NASA Open Science Data Repository (OSDR) now exposes sample metadata and processed assay data through a queryable biological API [1]. This makes it possible to assemble a cross-study cohort while retaining accession-level provenance. Public reference resources offer a second opportunity. ARCHS4 uniformly processes a large fraction of public human and mouse RNA-seq data [2], providing tissue-diverse reference profiles for pretraining models that would be underdetermined on OSDR alone.

Deep generative models can learn high-dimensional expression distributions. Conditional WGAN-GP models have reproduced tissue and cancer properties in GTEx and TCGA [3]. More recently, Lacan and colleagues adapted denoising diffusion probabilistic and implicit models to bulk transcriptomics and reported strong gene-correlation, neighborhood, adversarial, and downstream classification metrics [4]. GeneJEPA instead learns masked-gene representations without reconstructing expression [5]. These approaches solve different problems: a generator can sample expression, whereas a representation learner needs an additional decoder or generative objective before it can do so.

The model is only one part of the problem. Multi-study bulk RNA-seq can be represented as counts, CPM, TPM, or transformed and scaled expression; studies can be corrected, explicitly conditioned, modeled separately, or pooled. Published spaceflight workflows have used within-study standardization [24] and compared ComBat, ComBat-seq, and MBatch correction families [25]. MOBER offers a learned, inductive alternative based on an adversarial conditional variational autoencoder [26]. Any of these choices can improve one diagnostic while erasing flight-related structure or preserving study artifacts instead of biology.

We built a common framework around three model families and the preprocessing, harmonization, cohort, conditioning, and training choices surrounding them. Models were compared using correlation, neighborhood, adversarial, distributional, diversity, memorization, and FLT/GC-effect metrics. The OSDR-adapted DDIM was less distinguishable from real profiles, had lower distributional distance, and recovered accession-aware muscle effects while maintaining high fidelity, so it was used for downstream analysis.

Synthetic expression is commonly presented as a remedy for small sample size. Generated profiles, however, are not new biological replicates. After choosing diffusion for downstream analysis, we separated three questions: whether one pooled augmentation strategy helped at all, whether different tissues benefited from different synthetic-data uses, and whether prioritized genes were associated with flight in real samples.

Our primary biological question was whether tissue-specific analysis could reveal spaceflight responses obscured by multi-tissue pooling.

The tissue-specific analysis supplied its most coherent biological programs in thymus and soleus. Kidney, spleen, skin, pooled skeletal muscle, and adrenal gland supplied narrower candidates. Results from the remaining tissues define the exploratory boundary of the approach.

A Data sources

ARCHS4 mouse

997,515 profiles audited
17,244 selected across 20 tissues
Complete GEO-series splits

NASA OSDR API

1,610 biological profiles
75 accessions; FLT/GC labels
Study and material retained

Biological scope

Pooled multi-tissue generation
Tissue-specific analysis
Skeletal-muscle groups retained

B Configurable generative benchmark

Expression

Raw, CPM, TPM
log transforms
z-score, robust, MaxAbs

Harmonization

None; two study z-scores
ComBat/ComBat-seq
MBatch; MOBER

Generator

WGAN-GP; DDIM
GeneJEPA screened as
representation only

Training scope

OSDR only; ARCHS4 only
ARCHS4 pretrain plus
OSDR adaptation

Cohort structure

One or many studies
pooled or per tissue
accession balancing

Conditioning

FLT/GC; tissue; study
material; muscle group
and available covariates

C Generator metrics and model choice

WGAN-GP

Corr. 0.976; F1 0.985
AA 0.636; muscle recovery 0/6

DDIM

Corr. 0.974; F1 0.997
AA 0.475; muscle recovery 4/4

Downstream model

Lower AA and FD; muscle recovery 4/4
DDIM used for downstream analysis

AA near 0.5 indicates lower separability; generated profiles were never counted as additional animals.

D Five tissue-specific uses of generated expression

Training views

Real OSDR

Observed FLT/GC profiles

DDIM generated

Matched conditional profiles

Candidate arm fitted within each tissue

Real only

Rank: real
Fit: real

Generated only

Rank: generated
Fit: generated

Real + synth.

Rank: consensus
Fit: equal real/synth.

Guided; real fit

Rank: consensus
Fit: real

Guided; 5% synth.

Rank: consensus
Fit: real + 5% synth.

Within-study real-profile evaluation

BA, AUROC, AP on real holdouts
Profiles split within accession
Eligible arm selected per tissue

Selected arm -> repeated stable genes; FLT/GC effects and BH-FDR are calculated from real OSDR only

Figure 1. Configurable generative transcriptomics framework. (A) ARCHS4 supplied tissue-diverse reference profiles and the NASA OSDR API supplied flight and ground-control profiles with study provenance. (B) Configurable axes covered expression representation, harmonization, model family, training scope, cohort structure, and conditioning. (C) WGAN-GP and DDIM metrics were compared; DDIM's lower real-versus-synthetic separability and stronger accession-effect recovery supported its downstream use. (D) Each tissue compared real-only, generated-only, real-plus-generated, consensus-ranked real-only, and consensus-ranked low-weight synthetic training. Every arm was evaluated on held-out real profiles from represented accessions; biological effects and false-discovery rates came from real OSDR samples.

Materials and methods

Data sources

The OSDR Biological Data API was used to identify *Mus musculus* bulk RNA-seq assays with flight or ground-control labels [1]. Tissue and material names were harmonized while preserving study provenance. The resulting cohort contained 1,610 biological profiles from 75 accessions: 835 flight and 775 ground control. Full-transcriptome expression was converted to transcripts per million before selecting a 974-gene mouse landmark panel.

The local ARCHS4 mouse resource contained 997,515 public RNA-seq profiles [2]. All nine GEO series linked from the API-derived OSDR metadata were excluded before reference selection, removing 108 otherwise eligible profiles and preventing cross-resource sample overlap. A healthy-preferred, tissue-balanced subset of 17,244 profiles spanning 20 tissue classes was then used for model pretraining. Complete GEO series were assigned to one reference role, producing 10,150 training, 2,466 validation, and 4,628 test profiles with no series overlap. This backbone was used for both the held-out test and the broad development screens.

Table 1. Data scope.

Source	Profiles used	Biological scope	Role
ARCHS4 mouse v2.5	17,244	20 tissue classes	Mouse tissue pretraining
NASA OSDR API	1,610	75 accessions; 835 flight and 775 ground control	Spaceflight analysis

Configurable generative benchmark

The framework treated expression representation, feature space, harmonization, training source, cohort structure, conditioning, model family, and validation design as separate axes (Table 2). It was not an exhaustive Cartesian benchmark; lower-cost screens identified configurations for full training and repeat evaluation.

Table 2. Configurable pipeline and selected analysis branch.

Axis	Alternatives represented in the framework	Selected branch for downstream analysis
Expression	Raw, CPM, TPM; log1p/log2p1; z-score, robust, or MaxAbs scaling	Full-transcriptome TPM, then train-fitted MaxAbs scaling
Feature space	All shared genes, fold-selected HVGs, Reactome genes, mapped mouse L1000	974 mapped mouse L1000 landmarks
Harmonization	None, two study-wise z-score schemes, ComBat, ComBat-seq, three MBatch methods, MOBER	No global batch correction; accession represented as a condition
Training data	OSDR only, ARCHS4 only, ARCHS4 pretraining plus OSDR adaptation	ARCHS4 pretraining plus OSDR adaptation
Cohort structure	One or multiple studies; pooled or per-tissue fitting	All eligible OSDR accessions in a pooled tissue-conditioned generator
Conditions	FLT/GC, tissue, study, material, muscle group, and available design covariates	Tissue, FLT/GC, accession, and material
Model	WGAN-GP, DDIM, GeneJEPA representation screen	Factorized conditional DDIM
Validation	Grouped evaluation, repeat generation, unconditional controls	Four-seed 293-profile OSDR test

Preprocessing and harmonization

Each generator received its paper-based preprocessing and a shared benchmark representation. The WGAN-GP branch used log-transformed expression with training-gene z-scores [3]. The diffusion branch used TPM, a mapped mouse landmark panel, and training-fitted MaxAbs scaling [4]. GeneJEPA used its

sparse log-transformed token representation and a global nonzero-expression standardization [5]. All feature selection and transform statistics were fitted on training partitions.

Nine harmonization arms were compared in a matched liver experiment: no correction, two study-wise standardization schemes, ComBat, ComBat-seq, three MBatch methods, and MOBER [24-26]. Advancement required preservation of expression fidelity and FLT/GC effects, not simply reduced study separation. Methods without a natural frozen transform for a new batch were treated as transductive sensitivity analyses. Full transform definitions and results are provided in Supplementary Methods S7 and Supplementary Tables S13-S14.

Model training and selection

We implemented paper-based WGAN-GP and DDIM generators [3,4,12,13]. GeneJEPA was evaluated as a representation model but not as a generator because its released architecture has no expression decoder [5]. The DDIM was first trained on the 17,244-profile ARCHS4 reference, then adapted to OSDR with tissue, FLT/GC, accession, and material conditions. Exact architectures, optimization schedules, adaptation stages, seeds, and hardware records are provided in Supplementary Methods S4-S7.

Complete GEO series and OSDR accessions were grouped when study-level separation was required. Candidate generators were evaluated for correlation structure, neighborhood precision and recall, external real-versus-synthetic separability, distributional distance, diversity, memorization, and recovery of FLT/GC effects [14,15]. Metrics were assessed independently and each model's evaluation split was reported. Exact thresholds are provided in Supplementary Methods S6. The factorized DDIM had lower adversarial accuracy and distributional distance than WGAN-GP and recovered accession-aware muscle effects; it generated FLT or GC expression for represented tissue and study contexts and supplied the downstream screens.

Evaluation funnel and synthetic-guided analysis

Evaluation used three stages. First, we tested the most direct use of synthetic expression: pooled augmentation across all tissues. Individual tissue profiles were not averaged or collapsed; instead, they were combined in one FLT/GC classification problem governed by a common decision rule. The locked benchmark compared classifiers trained with real profiles, generated profiles, or real plus generated profiles. Second, because the pooled strategy did not improve performance, each tissue was evaluated separately with five candidate uses (Fig. 1D). Real-only ranking and fitting used observed OSDR profiles; generated-only ranking and fitting used DDIM profiles. Real-plus-generated training used consensus ranking and equal total real and synthetic weight. Both guided arms used real/synthetic consensus ranking: one fitted the classifier only on real profiles, while the other added condition-recentered generated profiles at 0.05 total synthetic weight. Nested development splits withheld profiles but retained representation from the same accessions. These are within-study development results. Synthetic attribution was retained only when the selected arm was nonworse than real-only training across balanced accuracy, AUROC, and average precision under the frozen eligibility rule.

Third, stable features from the selected tissue arm were compared with stable real-only features. Genes stable under both approaches were interpreted as reinforced. Genes stable only under the eligible synthetic-informed arm were interpreted as synthetic-promoted. "Synthetic-promoted" describes repeated feature selection; it does not mean that a gene was absent from real expression or biologically novel.

Flight-minus-ground effects were estimated within each OSDR study and then summarized with a random-effects model [16]. This kept mission-level contrasts separate before meta-analysis and prevented generated profiles from increasing the biological sample count.

Within each tissue, the 974 real-data gene-level meta-analysis P values were adjusted with the Benjamini-Hochberg procedure, and BH FDR below 0.05 defined the primary statistical inclusion rule [6]. Generated profiles were never entered as biological replicates. Synthetic selection status was recorded separately as reinforced, synthetic-promoted, real-only, or not stably selected. Accession-direction agreement, between-study heterogeneity, and leave-one-accession-out results were retained as

interpretation and sensitivity measures rather than inclusion requirements. The complete BH-FDR inventory is provided in Supplementary Table S11, its synthetic-guided subset in Table S10, and tissue-level counts in Table S12. Reactome was used to group selected genes into biological processes [7].

Table 3. Evaluation stages and permitted interpretation.

Stage	Analysis	Data separation	Question answered	Permitted interpretation
1	Pooled utility benchmark	Locked profiles from represented studies	Does one synthetic-data policy help across tissues?	Method-level positive or negative result; no biological claim
2	Tissue-specific five-arm screen	Held-out profiles within represented accessions	Which synthetic use, if any, helps each tissue?	Developmental utility and stable feature nomination
3	Real-data random-effects BH FDR	FLT-GC effects estimated separately within each accession	Are nominated genes associated with flight in observed studies?	Real biological association; synthetic status reported separately

Results

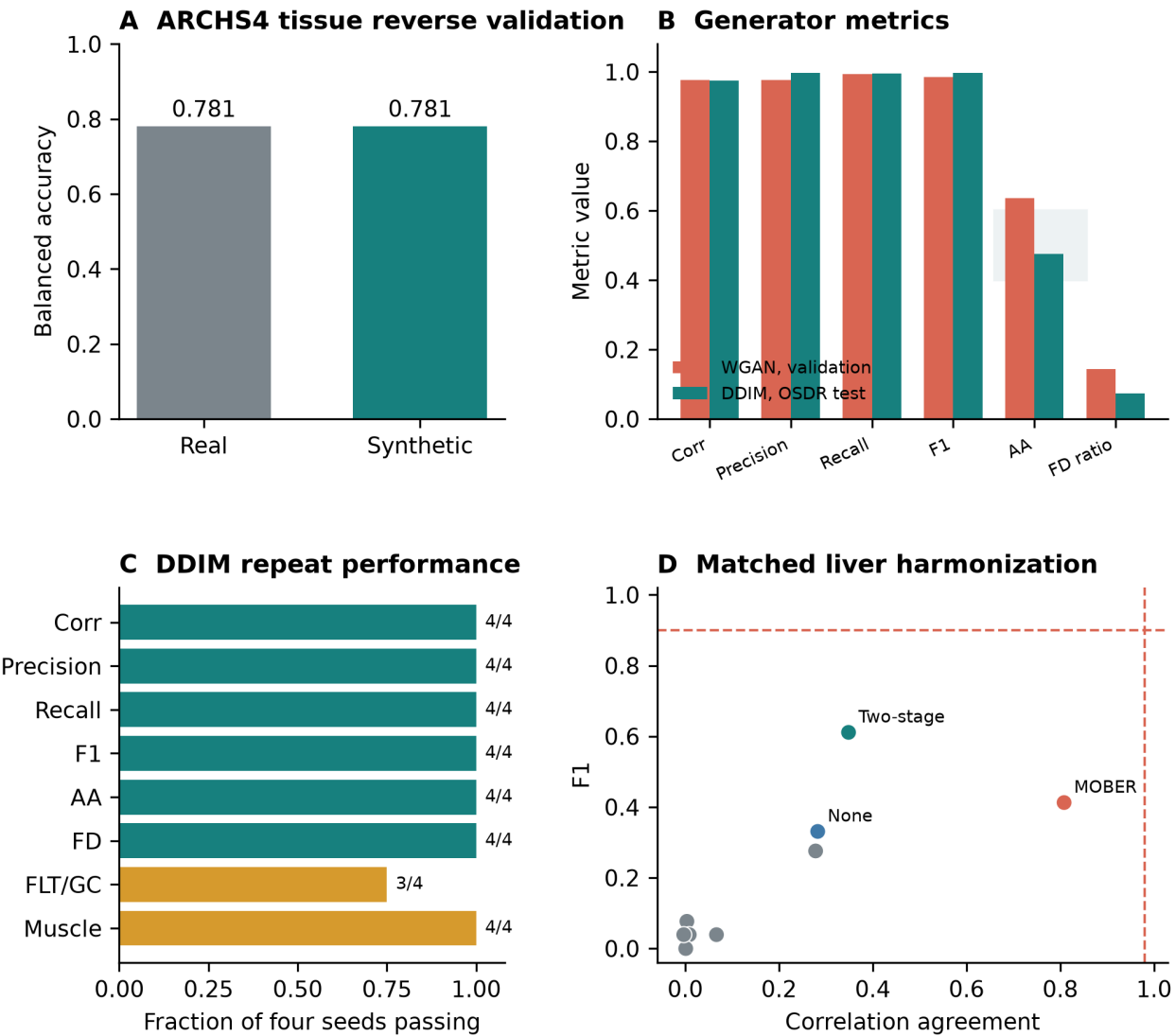
Generator metrics favored the OSDR-adapted diffusion model

The broad ARCHS4 DDIM retained tissue identity and passed most distributional tests on 4,628 held-out profiles, but its gene-correlation agreement missed the prespecified floor. It was therefore retained only as tissue-conditioned initialization. GeneJEPa also encoded tissue information, but its representation did not outperform expression directly and it could not generate profiles without an additional decoder.

No liver harmonization arm passed the joint fidelity and conditional-effect criteria (Fig. 2D; Supplementary Tables S13-S14). Some methods improved a single diagnostic while degrading neighborhood fidelity, real-versus-synthetic separability, or distributional distance. The selected path therefore retained TPM/MaxAbs expression without global correction and represented accession explicitly during OSDR adaptation.

WGAN-GP achieved correlation 0.976, F1 0.985, adversarial accuracy 0.636, and FD/real-P95 0.144; pooled FLT/GC recovery passed in 6/6 repeats, but accession-aware muscle recovery passed in 0/6. DDIM achieved correlation 0.974, F1 0.997, adversarial accuracy 0.475, and FD/real-P95 0.074; pooled recovery passed in 3/4 repeats and muscle recovery in 4/4. DDIM was therefore used downstream because it was less distinguishable from real data and recovered accession-aware condition effects while retaining high fidelity (Table 4).

Generator metrics support DDIM use in downstream analysis



WGAN values use validation data and DDIM values use the stated test data; metrics are not paired on one common split.

Figure 2. Generator metrics and model choice. (A) Real-trained and synthetic-trained classifiers retained the same broad ARCHS4 tissue balanced accuracy. (B) WGAN-GP validation metrics and DDIM test metrics on their stated evaluation splits; adversarial accuracy closer to 0.5 indicates lower real-versus-synthetic separability. (C) Fraction of four DDIM generation seeds passing each fidelity and effect-recovery criterion. (D) Correlation and F1 in the nine-arm matched liver harmonization benchmark. DDIM was used downstream because it combined near-chance adversarial accuracy, high fidelity, and accession-aware effect recovery.

Table 4. Generator metrics and model choice. Values are reported on each model's stated evaluation split and are not paired on one common split.

Model	Evaluation split	Corr.	Precision	Recall	F1	AA	FD/ real p95	FLT/GC recovery	Muscle recovery	Decision
Broad-reference DDIM	4,628 held-out ARCHS4 profiles	0.878	0.951	0.890	0.919	0.515	0.866	NA	NA	Initialization only
Study-conditioned WGAN-GP	536-profile validation; 6 seeds	0.976	0.976	0.994	0.985	0.636	0.144	6/6	0/6	Not used downstream
Factorized DDIM	293-profile OSDR test; 4 seeds	0.974	0.997	0.996	0.997	0.475	0.074	3/4	4/4	Used downstream

Diffusion generation recovered tissue-conditioned expression structure

The reverse trajectory provides a direct view of the model transforming noise into tissue-conditioned expression. In the ARCHS4 reference model, profiles moved from an overlapping noise cloud at timestep 1,000 toward the tissue-structured real-data manifold at timestep 0 (Fig. 3, top). After OSDR adaptation, locked real and generated profiles occupied similar tissue-defined regions, and flight and ground-control profiles remained interspersed within that broader structure (Fig. 3, bottom). These two-dimensional projections are descriptive: model selection used the full correlation, precision, recall, adversarial, Frechet-distance, and conditional-effect gates rather than visual similarity.

Pooled augmentation motivated tissue-specific analysis

We first asked whether synthetic profiles could simply expand one multi-tissue FLT/GC training cohort. On the OSDR test, balanced accuracy was 0.754 with real-only training, 0.695 with generated-only training, and 0.737 with real-plus-generated training. The corresponding AUROCs were 0.820, 0.751, and 0.791. Pooled augmentation therefore did not improve the broad classifier.

This result is consistent with strong tissue variation in bulk expression and with flight responses that differ among tissues. A common classifier can dilute tissue-specific condition signals even when the generator itself is conditioned on tissue. We therefore moved from a single pooled augmentation policy to separate tissue-level analyses. When synthetic use was selected separately by tissue, several arms improved balanced accuracy, AUROC, and average precision within represented studies (Table 5). Different tissues selected different uses: spleen, skin, and soleus selected real-plus-generated training; pooled skeletal muscle selected feature guidance with a real-only classifier; kidney and thymus selected low-weight guided training; and lung and adrenal gland selected generated-only training. These are development-screen results because profiles, rather than complete studies, were withheld.

Table 5. Selected tissue-specific development results. Every displayed synthetic-informed arm met the balanced-accuracy, AUROC, and average-precision eligibility rule. Complete canonical-tissue and muscle-group results are in Supplementary Tables S9 and S6.

Tissue	Selected synthetic use	Balanced accuracy, real to selected	AUROC, real to selected
Thymus	Low-weight guided training	0.733 to 0.844	0.818 to 0.887
Skeletal muscle, pooled	Feature-guided real-only classifier	0.861 to 0.932	0.924 to 0.961
Soleus	Real plus generated	0.925 to 0.963	0.980 to 0.980
Kidney	Low-weight guided training	0.654 to 0.707	0.680 to 0.771
Spleen	Real plus generated	0.517 to 0.648	0.540 to 0.704
Skin	Real plus generated	0.671 to 0.756	0.691 to 0.768
Lung	Generated only	0.664 to 0.742	0.680 to 0.830
Adrenal gland	Generated only	0.781 to 0.922	0.906 to 0.984

Upstream-parity DDIM generation across ARCHS4 mouse tissues

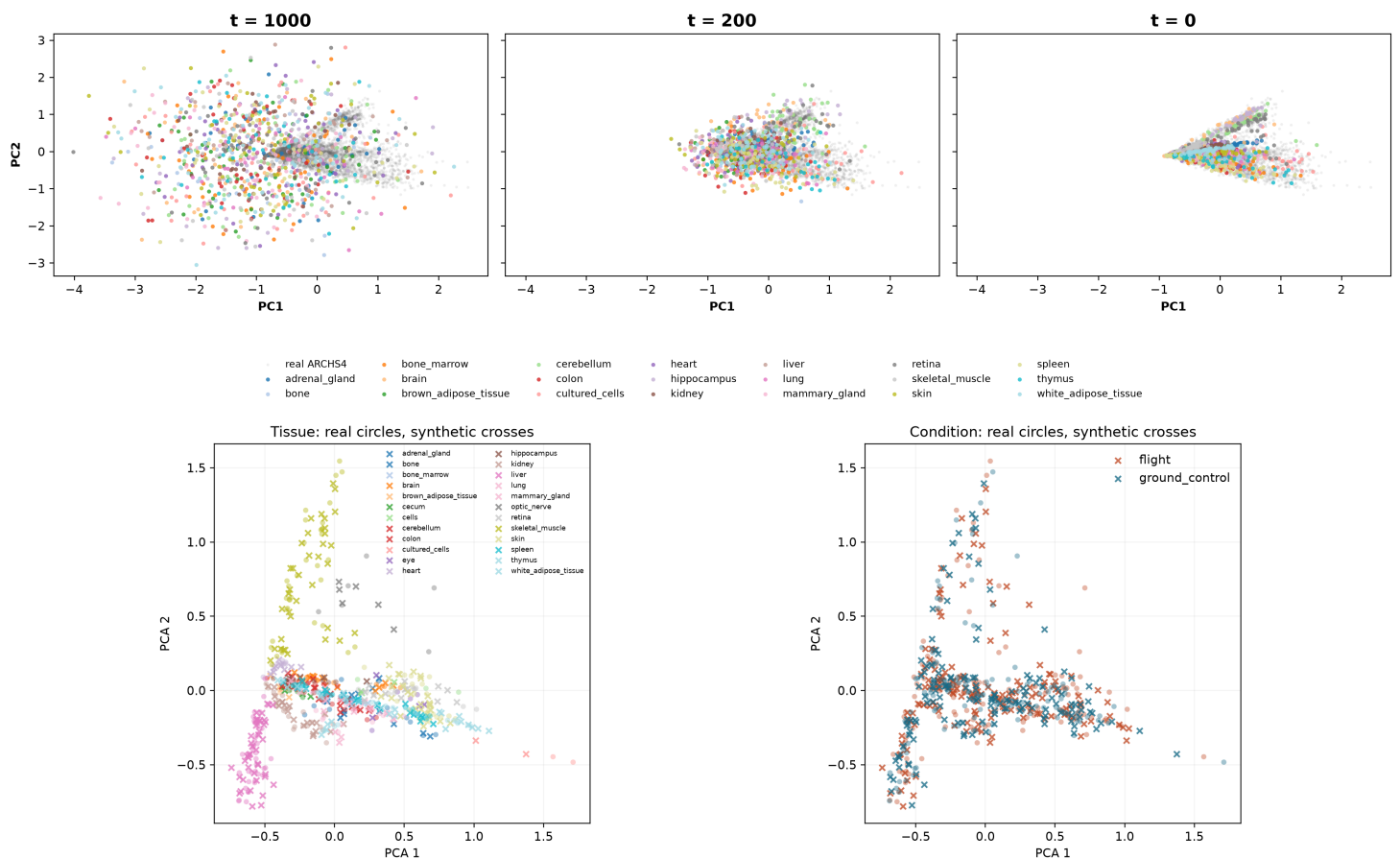


Figure 3. Diffusion generation across reference pretraining and OSDR adaptation. Top: ARCHS4 tissue-conditioned profiles at DDIM timesteps 1,000, 200, and 0 in a PCA space fitted to real reference expression; gray points are real ARCHS4 profiles and colors identify generated tissue conditions. Bottom: OSDR test profiles for generation seed 5020; circles denote real profiles and crosses denote generated profiles, colored by tissue on the left and flight condition on the right. PCA views are descriptive and do not replace quantitative validation metrics.

Synthetic-informed selection identified real-data associations

The real-data screen yielded 202 BH-FDR tissue-gene associations across 10 of 22 canonical tissue analyses and 257 across the five anatomical muscle groups. These are tissue-gene results rather than 459 unique or independent discoveries: a gene can be significant in more than one tissue, and pooled

skeletal muscle overlaps its anatomical subgroups. Forty-nine associations also entered stable synthetic-informed selection: 26 were synthetic-promoted and 23 were reinforced by both real-only and synthetic-informed selection.

All 459 P values and FDR values came from real profiles. Synthetic data affected only whether a gene was repeatedly prioritized and, for eligible augmentation arms, classifier fitting. Synthetic labels were suppressed where the selected arm failed its metric gate; quadriceps, EDL, and liver therefore contribute real-data associations but no synthetic-informed claims. The complete inventory, including associations with mixed study directions, is provided in Supplementary Tables S10-S12.

Synthetic-informed thymus analysis identifies lower proliferative renewal

The thymus screen selected low-weight synthetic-guided training. Balanced accuracy increased from 0.733 to 0.844, AUROC from 0.818 to 0.887, and average precision from 0.862 to 0.918 across repeated within-study splits. Sixteen thymus associations passed real-data BH FDR and entered stable synthetic-informed selection: 13 were synthetic-promoted and three were reinforced.

The strongest coherent subset contained flight-lower *Nusap1*, *Stmn1*, *Birc5*, *Cdk1*, *Top2a*, *Ccnb2*, *Aurka*, and *Ccne2*, all promoted by synthetic guidance, together with reinforced *Ube2c* and *Gmnn* (Fig. 4A). Each had a lower random-effects estimate across the five thymus accessions and BH FDR below 0.05. Reactome analysis of the synthetic-supported set identified mitotic cell cycle, DNA replication, S phase, APC/C regulation, and G2/M control (Fig. 4B).

These genes were present and statistically supported in the real data. Synthetic guidance changed their repeated selection behavior and organized them into a coherent panel; it did not create independent evidence or establish de novo biological discovery.

Prior thymus studies report related biology, although the present result is more specific. STS-135 mouse thymus showed changes in cell-cycle and DNA-damage programs, including lower checkpoint-related expression [8]. A later ISS experiment reported marked thymus mass loss and partial artificial-gravity rescue of cell-cycle expression [9]. The current signature emphasizes mitotic completion and replication rather than acute apoptosis alone. Agreement across accessions suggests a shared thymus response despite study heterogeneity, but it does not identify the responsible cell population.

Bulk thymus expression cannot distinguish lower transcription within proliferating thymocytes from loss or redistribution of proliferating cell populations. The defensible biological conclusion is lower abundance of a mitotic transcript program in flight, consistent with reduced proliferative renewal. Cell-resolved or histological confirmation is required to assign the effect to a cell-intrinsic mechanism.

Synthetic-informed thymus genes converge on a flight-lower mitotic program

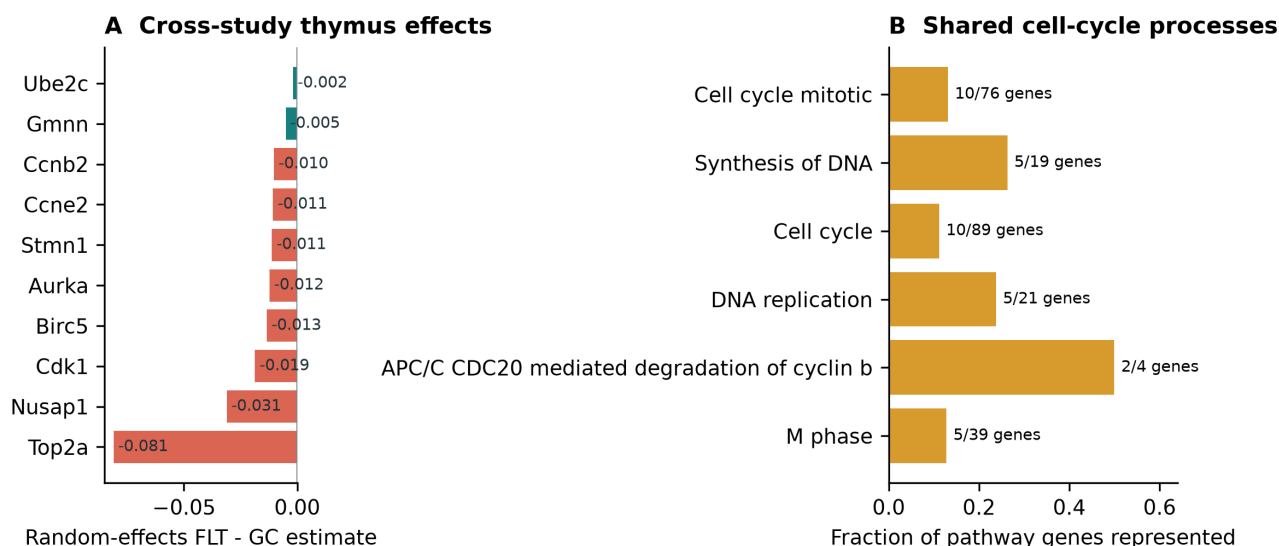


Figure 4. Thymus response to spaceflight. (A) Real-data random-effects flight-minus-ground estimates for ten genes across five thymus accessions; coral denotes synthetic-promoted genes and teal denotes reinforced genes. (B) The synthetic-supported set converges on mitotic and DNA-replication processes.

Anatomical separation exposes a soleus-specific metabolic program

Aggregate skeletal muscle concealed substantial anatomical heterogeneity. We therefore examined extensor digitorum longus, gastrocnemius, quadriceps, soleus, and tibialis anterior separately. Soleus produced the clearest biological pattern: its selected genes showed consistent flight effects across three accessions and converged on related metabolic processes.

The soleus screen selected real-plus-generated training. Balanced accuracy increased from 0.925 to 0.963, AUROC remained 0.980, and average precision increased from 0.980 to 0.986. Five BH-FDR genes were stable in both real-only and synthetic-guided selection: *Bdh1*, *Ech1*, *Bnip3*, and *Decr1* were lower in flight in all three accessions, while *Tpm1* was higher (Fig. 5B). Four genes passed leave-one-accession-out FDR; *Decr1* did not.

The model did not promote a soleus gene absent from stable real-only selection. Its contribution was reinforcement of a coherent existing pattern. Reactome connected the retained genes to mitochondrial protein turnover, mitochondrial fatty-acid beta-oxidation, and lipid metabolism (Fig. 5C). *Bdh1* and *Ech1* support oxidative substrate handling, *Bnip3* supports mitochondrial quality control, *Decr1* contributes to unsaturated fatty-acid oxidation, and *Tpm1* suggests contractile remodeling.

Prior 30-day spaceflight profiling of mouse soleus reported a slow-to-fast shift and broad changes in oxidative metabolism, PPAR signaling, and contractile genes [10]. Unloading studies have also reported reduced soleus fatty-acid oxidation [11]. This is literature-supported panel reinforcement rather than de novo gene discovery.

Unlike thymus, soleus was represented during model development. Its cross-study consistency makes it a focused biological hypothesis, but an entirely unseen soleus study is still needed for independent confirmation.

Anatomical separation reveals a soleus-specific oxidative-metabolism hypothesis

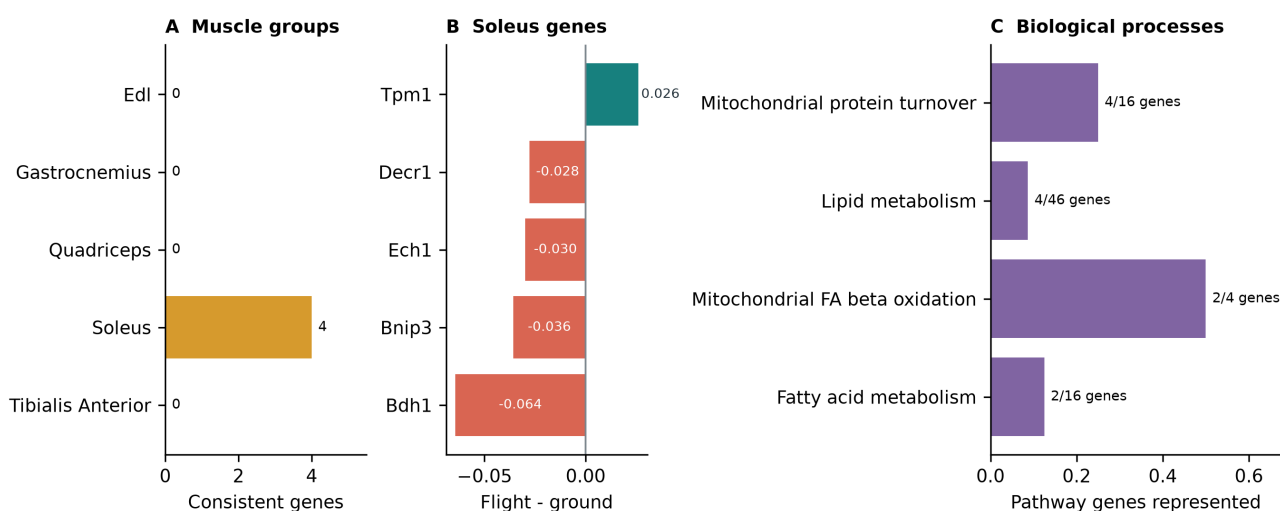


Figure 5. Skeletal-muscle and soleus response. (A) Number of synthetic-informed genes that also pass real leave-one-accession-out FDR in each anatomical muscle group. (B) Five reinforced soleus BH-FDR genes with consistent real flight effects. (C) Their strongest shared biological processes center on mitochondrial turnover and lipid metabolism.

Other muscle groups provide narrower hypotheses

The pooled skeletal-muscle screen selected synthetic-guided feature ranking and improved balanced accuracy, AUROC, and average precision by 0.071, 0.036, and 0.037. Twelve BH-FDR genes were synthetic-informed, nine of which passed leave-one-accession-out sensitivity. The associated gene sets

included interferon signaling and sialic-acid metabolism. Because pooled muscle combines anatomical groups with distinct physiology, this result is interpreted alongside, rather than instead of, the soleus analysis.

Among the remaining anatomical groups, tibialis anterior retained a real-plus-generated arm and gastrocnemius retained a low-weight guided arm. Tibialis anterior reinforced *Cdkn1a*, *St3gal5*, and *Bnip3* and promoted *Cebpd*; gastrocnemius promoted *Fhl2* and *Nfkbia*. None of these genes passed leave-one-accession-out FDR. EDL and quadriceps selected real-only arms, so their BH-FDR genes remain in the complete real-data inventory but are not attributed to the synthetic workflow.

Spleen screen nominates adhesion and cytoskeletal genes

The spleen screen selected real-plus-generated training. Balanced accuracy increased from 0.517 to 0.648, AUROC from 0.540 to 0.704, and average precision from 0.589 to 0.749. Synthetic guidance promoted flight-higher *Rai14*, *Ptprk*, and *Myf9* and reinforced flight-higher *Lox1l*. *Rai14*, *Ptprk*, and *Lox1l* had the same direction in all six studies; *Myf9* agreed in five of six. None passed leave-one-accession-out FDR, and the set did not produce significant Reactome enrichment.

The four genes provide a tentative structural hypothesis rather than a pathway claim. RAI14 links actin-associated mechanosensing to Hippo signaling, PTPRK regulates cell-cell junctions, MYL9 contributes to actomyosin contractility, and LOXL1 supports elastic extracellular-matrix maintenance [21-23]. Their shared direction is compatible with altered adhesion, mechanics, or tissue architecture, but the present bulk data cannot identify a common source cell or establish one mechanism.

Other organ responses were heterogeneous

Lung produced large repeated development-screen gains and cell-cycle, senescence, and PI3K/AKT-related enrichment, but no selected gene passed the real-data BH-FDR screen. We therefore treat the lung result as predictive development evidence without a supported gene-level biological claim.

Skin selected real-plus-generated training and promoted flight-higher *Plscr1*. *Plscr1* passed BH FDR and had the same direction in all six studies, although it did not pass leave-one-accession-out FDR. Broader selected sets were enriched for G1/S and DNA-repair processes, consistent with prior OSDR skin analyses reporting strong mission, strain, recovery-interval, and anatomical-site dependence [17]. This is a literature-aligned developmental candidate rather than a confirmed transferable signal.

Kidney produced the clearest secondary gene-level result. *Slc37a4* was reinforced by both selection arms and was higher in flight in all six kidney accessions. Synthetic guidance also promoted flight-higher *Inpp4b*; it passed BH FDR and remained significant in leave-one-accession-out analysis, although one of six accession effects was negative. The selected pair did not yield significant Reactome enrichment. *Slc37a4* supports renal glucose handling, while *Inpp4b* nominates phosphoinositide signaling. Both require independent study confirmation and should be interpreted alongside prior reports of strain-dependent lipid, ECM, TGF-beta, Wnt, and nephron-remodeling responses [18-20].

Adrenal gland produced flight-lower synthetic-promoted *Psmb8* and reinforced *Tspan4*, each with the same direction in all three studies but neither passing leave-one-accession-out FDR. Retina showed repeated predictive gains and selected-set enrichment but no synthetic-informed BH-FDR gene. Liver selected the real-only arm: its 19 real-data BH-FDR genes remain in the complete inventory, but none support a synthetic-guided claim. Quadriceps and EDL likewise selected real-only arms.

Synthetic-guided development and real-data support identify tissue priorities



Figure 6. Tissue evidence. (A) Repeated development-screen changes relative to real-only models for selected tissues. (B) BH-FDR genes promoted or reinforced by the synthetic workflow; synthetic labels are suppressed where the generated arm failed its metric gate. (C) Thymus and soleus provide the clearest process-level hypotheses, kidney supplies a focused secondary pair, and spleen, skin, and adrenal gland remain exploratory.

Table 6. Selected synthetic-guided biological interpretations by tissue. Complete real-data BH-FDR results are in Supplementary Tables S11-S12.

Tissue	Main signal	Interpretation
Thymus	Lower mitotic genes; APC/C, G2/M, and DNA replication	Coherent synthetic-informed panel supported by real-data BH FDR
Soleus	Reinforced FLT-lower <i>Bdh1</i> , <i>Ech1</i> , <i>Bnip3</i> , and <i>Decr1</i> ; FLT-higher <i>Tpm1</i>	Coherent mitochondrial and lipid-metabolism program; no promoted gene
Skeletal muscle, pooled	12 synthetic-informed BH-FDR genes; interferon and sialic-acid terms	Developmental complement to the anatomically specific soleus result
Kidney	Promoted <i>Inpp4b</i> and reinforced <i>Slc37a4</i> , both FLT-higher	Focused renal metabolic-signaling hypothesis
Spleen	Promoted <i>Rai14</i> , <i>Ptprk</i> , and <i>Myl9</i> ; reinforced <i>Lox11</i>	Developmental adhesion/cytoskeletal hypothesis
Skin	Promoted FLT-higher <i>Plscr1</i> plus cell-cycle/DNA-repair enrichment	Literature-aligned developmental candidate
Adrenal gland	Promoted <i>Psm8</i> and reinforced <i>Tspan4</i> , both FLT-lower	Three-study developmental candidate
Tibialis anterior	Reinforced <i>Cdkn1a</i> , <i>St3gal5</i> , and <i>Bnip3</i> ; promoted <i>Cebpd</i>	Exploratory stress and metabolic response
Gastrocnemius	Promoted <i>Fhl2</i> and <i>Nfkb1a</i>	Exploratory two-gene result without a coherent pathway
Lung	Generated-only development improved all three predictive metrics	Predictive development result without a BH-FDR gene in the 974-gene panel
Retina	Predictive and pathway gains without a synthetic-informed BH-FDR gene	Exploratory gene/pathway mismatch
Quadriceps, EDL, liver	Screen retained real-only arms	Real-data BH-FDR associations are not synthetic-guided findings

Discussion

Generator metrics favored diffusion for downstream analysis

Both WGAN-GP and DDIM reproduced expression structure, but DDIM was less distinguishable from real profiles and recovered accession-aware muscle effects. OSDR adaptation was needed because the broad ARCHS4 model retained tissue identity but missed the correlation target. Harmonization also showed that reducing study structure could damage expression neighborhoods or FLT/GC effects, so accession was represented explicitly. These results support using DDIM for this analysis; they do not establish universal superiority over adversarial training.

What synthetic data added

Pooled augmentation did not improve FLT/GC classification, consistent with organ-specific responses being obscured by one decision rule. Tissue-specific analysis was more informative, but no single synthetic-data use worked across organs. Depending on the tissue, generated profiles contributed through direct augmentation, low-weight regularization, or feature ranking.

These results support synthetic expression as a regularizer or feature prior, not as additional biological sample size. All associations were tested in observed profiles after estimating FLT/GC effects separately by accession. A reinforced gene was selected with and without synthetic guidance; a promoted gene was selected stably only with guidance. Neither label establishes novelty or replaces validation in an independent experiment.

Thymus and soleus define the clearest biological hypotheses

The thymus result refines established spaceflight immune biology. Prior work documents thymic involution and altered cell-cycle expression [8,9]. The current signature concentrates on cyclins, CDK1, UBE2C, BIRC5, NUSAP1, geminin, APC/C-mediated protein turnover, and G2/M control. Together these support lower proliferative renewal or a lower proportion of cycling thymocytes. Because the data are bulk, composition and cell-intrinsic regulation remain inseparable.

The soleus result addresses a different physiological axis. Weight-bearing slow muscle is especially sensitive to unloading. Lower *Bdh1*, *Ech1*, *Bnip3*, and *Decr1* together with higher *Tpm1* describe reduced oxidative substrate handling, altered mitochondrial quality control, and contractile remodeling. This is compatible with known soleus atrophy, altered lipid metabolism, and slow-to-fast transition [10,11]. Synthetic guidance reinforced this compact fatty-acid-oxidation panel across soleus studies.

Secondary and negative tissue results

The other tissues define the method's exploratory boundary. Kidney supplied a focused two-gene hypothesis, while spleen, skin, and adrenal gland supplied narrower candidates without comparable pathway coherence. Retina and lung improved in development without a synthetic-informed BH-FDR gene, and liver, quadriceps, and EDL selected real-only arms. Follow-up should prioritize the thymus mitotic panel and soleus metabolic panel, then test kidney *Inpp4b* and *Slc37a4* and the spleen structural candidates.

Limitations

The tissue screens withheld profiles within represented accessions, so they measure development utility rather than transfer to a new mission, strain, or processing protocol. The configurable matrix was a gated search rather than a full factorial benchmark, and the selected path should not be interpreted as a universal model optimum.

The 974-gene landmark panel excludes potentially relevant genes. FDR was controlled within each tissue family, and heterogeneous study effects weaken a common-response interpretation even when the pooled association is significant. Bulk data also cannot separate cell-intrinsic regulation from changes in cell composition. Finally, the generator represents only tissue, condition, accession, and material contexts seen during adaptation. Independent studies and targeted experiments are required to confirm the proposed programs; detailed sensitivity analyses and safeguards are provided in the supplement.

Conclusions

Synthetic expression was useful only when its role was selected for the tissue and evaluated against real-only training. A pooled augmentation policy failed. Thymus and soleus provide the clearest process-level hypotheses, while kidney *Inpp4b* and *Slc37a4* remain focused secondary candidates.

Skin, spleen, pooled skeletal muscle, adrenal gland, gastrocnemius, and tibialis anterior provide narrower developmental hypotheses. Lung and retina improved during development but lacked a corresponding synthetic-informed BH-FDR gene in the landmark panel. Quadriceps, EDL, and liver do not support synthetic-guided claims because their real-only arms were retained. Independent biological data are needed to test all of these hypotheses.

Data and code availability

OSDR data were accessed through the public Biological Data API [1]. ARCHS4 and Reactome are public resources [2,7]. Repository scripts are under `src/nasa_mouse_rna_diffusion/`; frozen run outputs are under `outputs/generative_benchmark/`; this paper package is under `paper/synthetic_guided_spaceflight/`. The figure and table builder is `nasa_mouse_rna_diffusion.build_synthetic_guided_paper`. Public repository URL and archival DOI will be added after author review.

Ethics statement

This study is a secondary computational analysis of publicly available animal-study data. No new animals were used.

Competing interests

The author declares no competing interests.

Acknowledgments

The author acknowledges the NASA Open Science Data Repository, GeneLab data-processing teams, original flight-study investigators, ARCHS4, Reactome, and the developers of the evaluated generative-model implementations. Program and mentor acknowledgments require final author review.

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